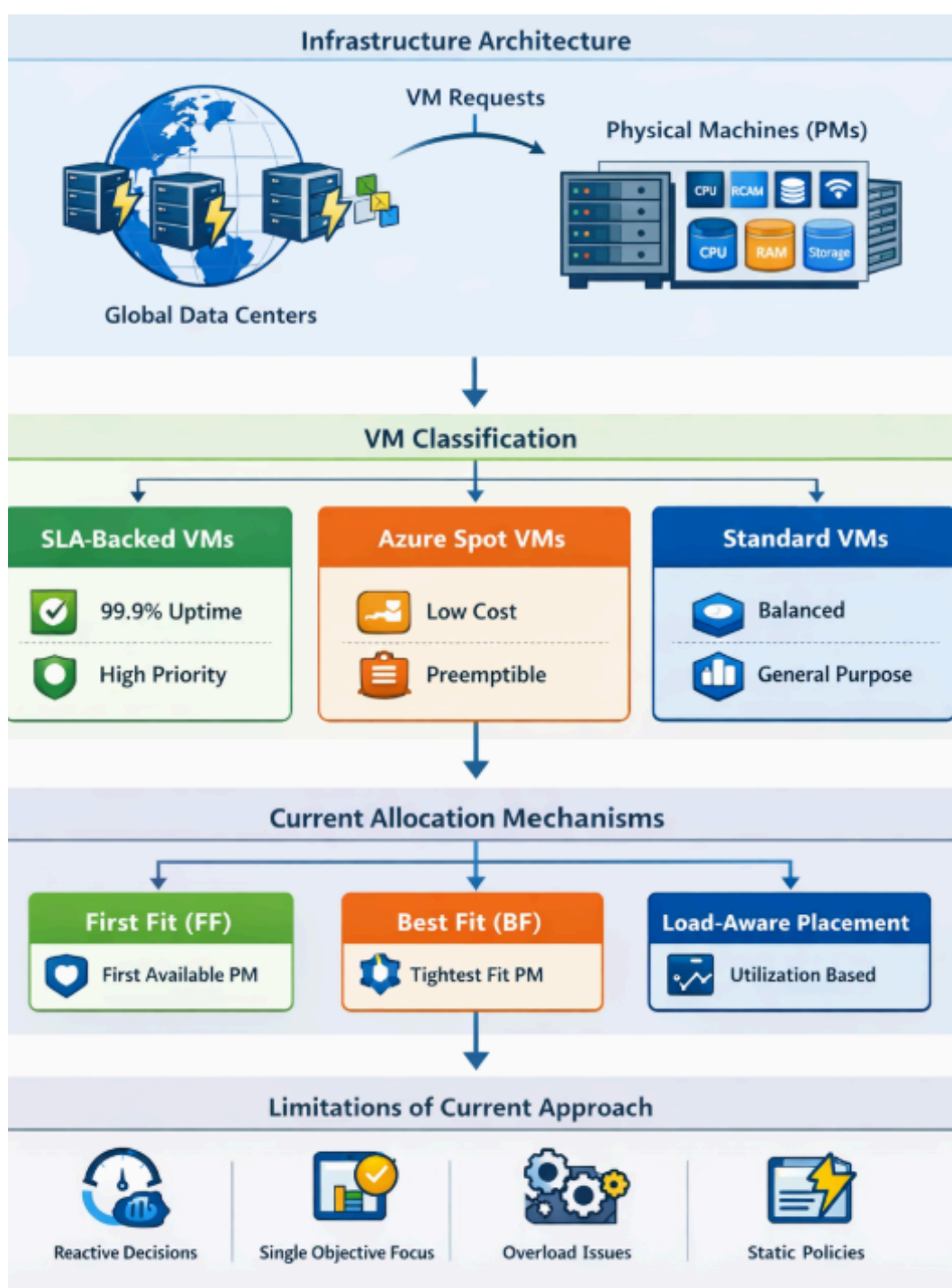


Introduction

Microsoft, founded in 1975 in the United States by Bill Gates and Paul Allen, is a global technology leader operating in software, hardware, and cloud computing. With products such as Windows, Office, Azure, and Xbox, it serves both individual and corporate users worldwide. Azure, supported by a global network of data centers, delivers scalable, reliable, and cost-effective cloud solutions and plays a key role in digital transformation. This project aims to optimize resource management and capacity planning for Azure cloud services using mathematical modeling, focusing on efficient resource utilization, service continuity, and improved operational efficiency.

Current System

Microsoft Azure operates a global network of virtualized data centers hosting multiple types of virtual machines: SLA-backed VMs with high availability guarantees, Azure Spot VMs using excess capacity at lower cost, and Standard VMs for general-purpose use. Currently, VM placement relies on heuristic algorithms such as First Fit and Best Fit, which are fast and scalable but focus mainly on minimizing active physical machines. This single-objective approach leads to limitations, including reactive placement, poor handling of multiple objectives, suboptimal admission decisions, and limited adaptability to dynamic workloads. These challenges highlight the need for a more optimization-driven approach to VM placement and admission control.



Motivation

Cloud computing is a key part of modern digital infrastructure, with Microsoft Azure serving millions of enterprise customers worldwide. However, large-scale operations face major challenges in efficient resource management. These include low data center utilization and high energy costs, strict SLA requirements that demand reliable prioritization, and QoS constraints that require keeping machine utilization below safe limits. To address these competing objectives, this study proposes a mathematical optimization framework that integrates load balancing, priority-based admission control, and capacity utilization while ensuring QoS.

Problem Definition

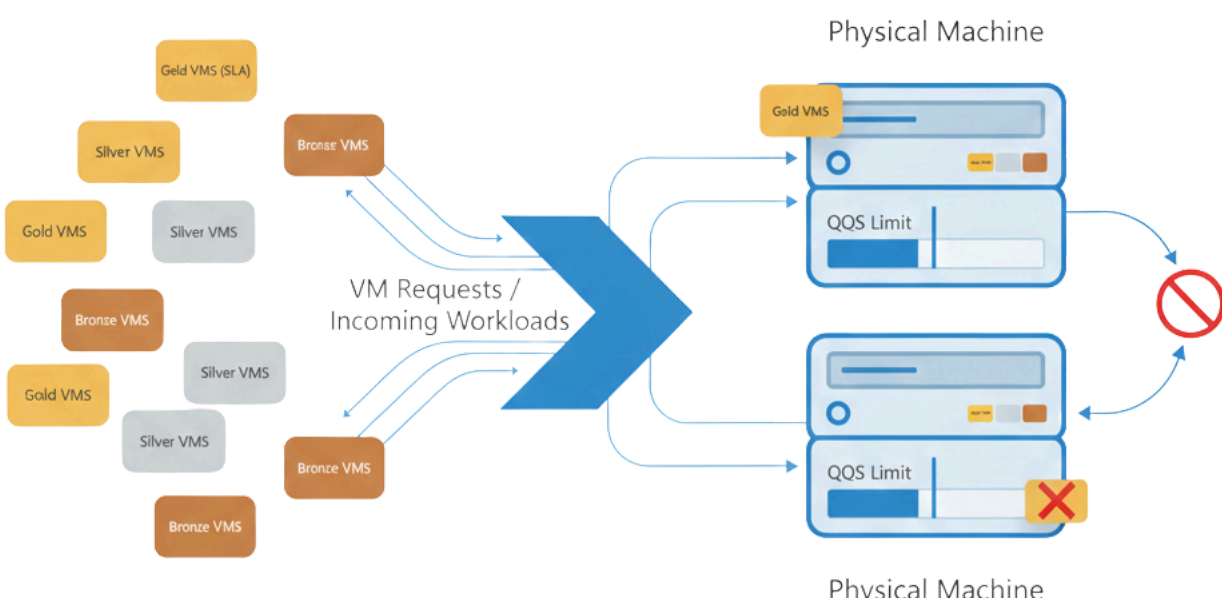
- Each VM requests a certain amount of CPU

- Each PM has a finite capacity and an upper QoS utilization limit

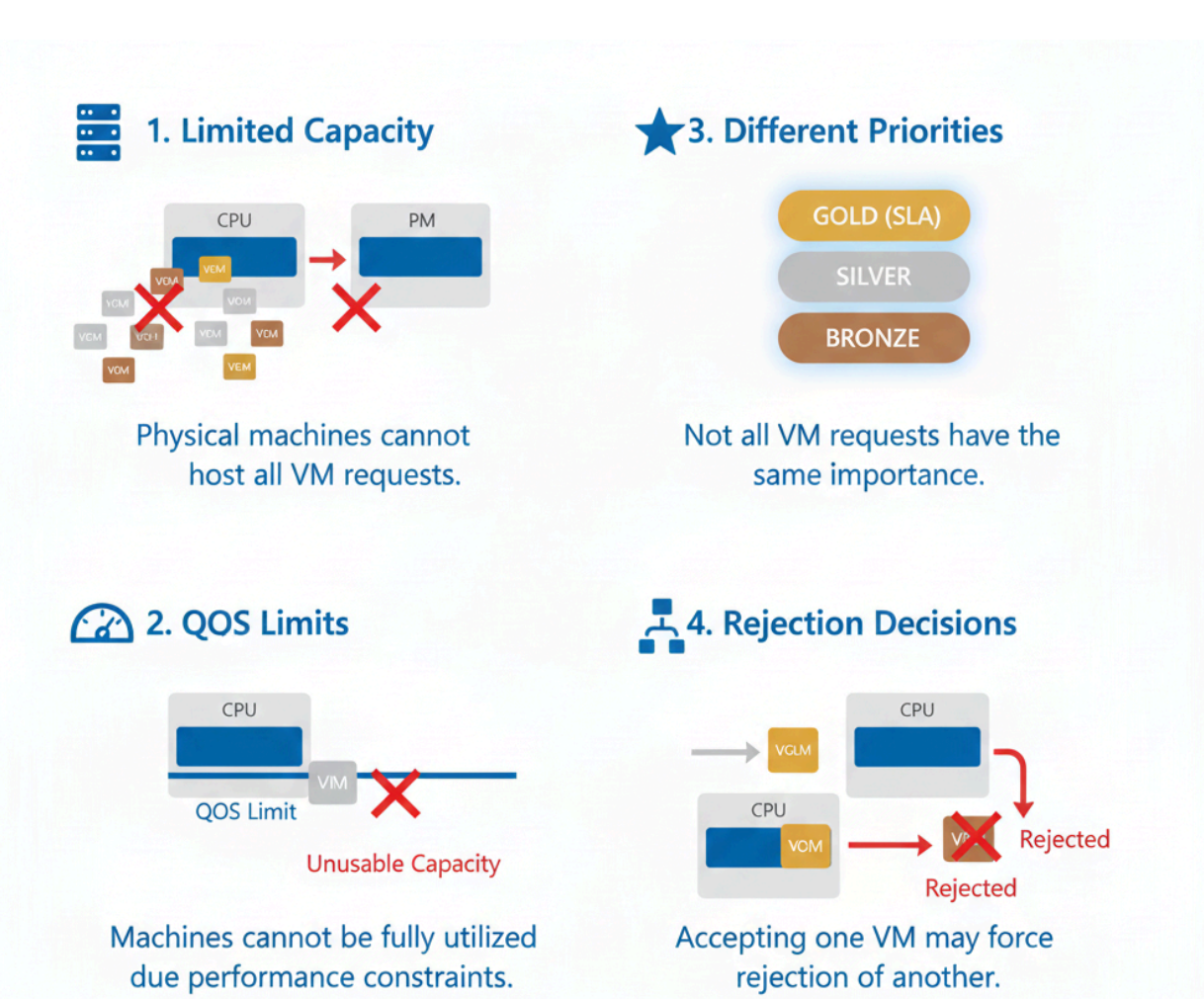
- VM requests can be accepted or rejected (admission control)

- Some VMs are mandatory (SLA) and cannot be rejected

- The goal is to make balanced, efficient, and priority-aware decisions



Why Is VM Placement Challenging?



Different Service Levels



Mathematical Model

Sets

$P = \{p_1, p_2, p_3\}$ (physical machines (PM))

$V = \{v_1, \dots, v_{13}\}$ (virtual machines (VM))

Parameters

$C_p > 0$ (PM p 's CPU capacity)

$\rho_p^0 \in [0, 1]$ (PM p 's initial usage rate)

$L_p^0 = \rho_p^0 C_p$ (PM p 's initial load)

$d_v > 0$ (CPU demand of VM v)

$w_v > 0$ (Priority/weight coefficient of VM v)

$\text{mandatory}_v \in \{0, 1\}$ (VM v is not rejected? 1 is SLA)

$U_{\max} \in (0, 1]$ (QoS usage limit)

$W_{\max} = \sum_{v \in V} w_v$ (upper limit for weighted rejection)

$D_{\max} = \sum_{v \in V} d_v$ (total demand ceiling)

Decision Variables

$x_{vp} \in \{0, 1\}$ (If VM is assigned to PM, then 1)

$u_v \in \{0, 1\}$ (If the VM is verified 1)

$\text{UsedCPU} \geq 0$ (Total CPU usage of accepted VMs)

$z_1 \in [0, 1]$ (normalized maximum utilization metric)

$z_2 \in [0, 1]$ (normal maximum utilization metric)

$z_3 \in [0, 1]$ (unused CPU metric normalized)

$Z \in [0, 1]$ (final normalized target value)

The model includes constraints that ensure each virtual machine is either assigned or rejected, prohibit the rejection of SLA-backed virtual machines, limit physical machine utilization under QoS thresholds, and properly define the normalized performance metrics within a mixed integer programming framework.

Constraints

Assignment & SLA Constraints

$$\sum_{p \in P} x_{vp} + u_v = 1 \quad \forall v \in V$$

- Each virtual machine is either assigned to exactly one physical machine or rejected
- Service Level Agreements (SLAs) are strictly enforced and non-negotiable

Capacity & QoS Constraints

$$L_p^0 + \sum_{v \in V} d_v x_{vp} \leq u_{\max} C_p \quad \forall p \in P$$

- Physical machine capacity limits are respected

- Quality-of-Service-aware resource allocation

$$\text{UsedCPU} = \sum_{v \in V} \sum_{p \in P} d_v x_{vp}$$

- Total CPU consumption across all physical machines

Objective Function

$$\text{Minimize } Z = \frac{z_1 + z_2 + z_3}{3}$$

- Balanced optimization of load distribution, rejection minimization, and resource utilization

Worst-case PM load

$$\frac{L_p^0 + \sum_{v \in V} d_v x_{vp}}{U_{\max} C_p} \leq z_1 \quad \forall p \in P$$

- Normalized load of the most loaded physical machine

Weighted VM Rejection

$$\frac{\sum_{v \in V} w_v u_v}{W_{\max}} \leq z_2$$

- Normalized weighted virtual machine rejection

Unused CPU Ratio

$$1 - \frac{\text{UsedCPU}}{D_{\max}} \leq z_3$$

- Normalized unused CPU capacity

VM CPU Demand Scenarios

Scenario 1: Overloading Scenario

Total VM demand exceeds physical capacity

- Used to evaluate rejection and SLA enforcement under overload conditions

VM Cpu Demands(overload)
d("v1") = 500;
d("v2") = 5000;
d("v3") = 500;
d("v4") = 500;
d("v5") = 500;
d("v6") = 1000;
d("v7") = 500;
d("v8") = 5000;
d("v9") = 1000;
d("v10") = 2000;
d("v11") = 500;

Scenario 2: Below Capacity Scenario

Total VM demand is lower than system capacity

- Enables analysis of balanced allocation without forced rejection

VM Cpu Demands(Lower)
d("v1") = 500;
d("v2") = 500;
d("v3") = 500;
d("v4") = 500;
d("v5") = 500;
d("v6") = 1000;
d("v7") = 500;
d("v8") = 500;
d("v9") = 1000;
d("v10") = 2000;
d("v11") = 500;

Scenario 3: Full Capacity Scenario

Total VM demand equals total physical capacity

- Tests optimal load distribution with zero slack capacity

VM Cpu Demand (Full Capacity)
d("v1") = 500;
d("v2") = 2000;
d("v3") = 500;
d("v4") = 1000;
d("v5") = 500;
d("v6") = 1000;
d("v7") = 1500;
d("v8") = 500;
d("v9") = 1000;
d("v10") = 500;
d("v11") = 1000;

Results

In the current system, virtual machine admission decisions are made under limited CPU capacity and strict QoS constraints.

After applying the proposed QoS-aware VM placement model with admission control, all SLA-protected virtual machines are successfully assigned, while low and medium priority requests are selectively rejected only under overload conditions. The model achieves balanced utilization across all physical machines without violating QoS limits.

